

TAMPERE PIRKKALA, Finland

WMO: 029440

Lat: 61.420N

Lon: 23.617E

Elev: 358

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.9	-5.5	-15.2	2.4	-10.5	-9.4	3.4	-5.1	18.9	29.6	17.0	29.6	2.1	20	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	80.6	63.8	77.1	62.5	73.7	61.2	67.1	75.7	65.0	73.4	63.1	71.0	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	91.1	70.6	61.8	83.7	68.4	59.7	77.5	66.1	31.8	75.7	30.1	73.6	28.7	71.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	15.4	13.8	DB	-16.0	83.9	7.0	3.3	-21.1	86.3	-25.2	88.2	-29.1	90.1	-34.3	92.5	(4)
(5)			WB	-16.4	69.4	7.0	2.6	-21.4	71.3	-25.5	72.8	-29.4	74.3	-34.5	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	20.7	20.3	26.9	38.4	48.7	56.8	62.4	59.6	50.7	40.3	32.2	25.6		
	DBStd	16.77	11.57	12.04	8.54	6.89	7.38	6.20	5.57	5.55	6.20	7.10	7.94	10.79		
	HDD50	4596	908	831	715	354	114	10	0	2	66	306	534	756		
	HDD65	9070	1373	1251	1180	799	508	257	119	183	429	766	984	1221		
	CDD50	1072	0	0	0	5	73	214	386	301	87	6	0	0		
	CDD65	67	0	0	0	0	2	10	39	16	0	0	0	0		
	CDH74	685	0	0	0	0	37	124	392	132	0	0	0	0		
	CDH80	108	0	0	0	0	3	16	78	11	0	0	0	0		
	WSAvg	6.4	6.6	6.4	6.5	6.3	6.5	6.4	5.8	5.6	6.0	6.6	6.8	7.0		
	PrecAvg	22.3	1.4	1.0	1.1	1.3	1.4	2.0	2.8	2.9	2.4	2.2	2.1	1.7		
(15) Precipitation	PrecMax	34.0	2.9	2.6	2.4	3.5	3.5	6.7	5.2	7.2	4.4	5.1	3.8	3.6		
	PrecMin	15.8	0.5	0.2	0.0	0.0	0.3	0.6	0.8	0.9	0.2	0.5	0.4	0.3		
	PrecStd	3.8	0.6	0.6	0.6	0.8	0.7	1.3	1.3	1.5	1.0	1.1	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	41.2	51.1	66.9	78.6	82.3	86.0	81.4	70.7	57.5	49.2	45.1		
		MCWB	38.8	37.7	41.7	51.4	60.0	63.8	66.7	64.1	60.3	53.1	48.1	44.0		
	2%	DB	37.4	37.7	44.6	60.4	73.2	77.4	82.2	77.4	66.6	54.0	46.6	42.5		
		MCWB	36.6	36.2	38.3	47.9	57.2	61.0	65.9	63.3	58.3	51.5	45.6	41.9		
	5%	DB	35.5	35.8	40.8	55.5	68.2	73.5	78.6	74.5	63.1	52.3	44.8	39.6		
		MCWB	35.0	34.7	36.2	45.6	54.5	59.3	64.0	62.3	56.5	50.3	43.9	38.8		
	10%	DB	33.8	34.0	37.5	51.4	63.5	69.8	74.9	71.1	60.6	50.4	42.6	37.5		
		MCWB	33.2	33.2	34.3	42.9	51.8	57.9	63.0	60.9	55.8	48.7	41.9	36.7		
	0.4%	WB	39.3	38.4	43.1	53.4	62.7	67.3	70.5	68.6	62.0	54.4	48.4	44.4		
		MCDB	39.5	39.7	48.5	63.4	73.5	76.6	80.0	76.8	67.5	56.5	49.0	44.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	36.5	39.1	49.6	59.2	64.0	68.0	65.7	59.9	52.6	45.8	41.7		
		MCDB	37.0	37.4	43.1	58.1	68.8	73.2	77.0	72.8	64.5	54.1	46.3	42.2		
	5%	WB	34.8	34.8	36.8	46.6	56.3	61.4	66.1	63.7	58.1	50.9	43.9	39.0		
		MCDB	35.2	35.6	39.6	53.9	65.9	70.3	74.8	71.1	62.0	52.3	44.5	39.5		
	10%	WB	33.2	33.3	34.9	43.8	53.1	59.0	64.0	62.0	56.4	48.9	41.9	36.8		
		MCDB	33.6	34.0	37.0	50.4	62.2	67.5	72.6	68.8	59.8	50.2	42.5	37.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.4	14.7	16.9	19.3	17.7	17.3	17.1	14.7	10.1	7.3	8.7		
		MCDBR	7.0	9.5	17.4	24.8	27.1	24.5	23.5	22.2	18.5	11.0	7.8	7.2		
		MCWBR	6.8	8.6	12.1	14.5	13.6	11.8	10.4	10.7	11.4	8.6	7.5	6.9		
	5% WB	MCDBR	7.0	8.9	14.8	22.5	22.9	20.6	19.6	18.6	15.6	9.7	7.5	7.0		
		MCWBR	6.9	8.3	10.9	13.8	12.8	11.2	9.8	9.9	10.4	8.2	7.4	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.295	0.321	0.357	0.353	0.349	0.374	0.360	0.336	0.321	0.274	0.358		
	taud		2.089	2.171	2.258	2.368	2.441	2.474	2.423	2.452	2.516	2.413	2.186	2.604		
	Ebn at Noon		144	212	251	265	276	279	267	262	249	206	135	96		
	Edh at Noon		12	22	29	31	31	31	32	29	23	18	11	8		
(44) All-Sky Solar Radiation	RadAvg		69	248	716	1170	1593	1695	1597	1251	746	317	82	39		
	RadStd		13	56	89	133	169	131	161	138	95	61	16	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	N/A	N/A	N/A	-266	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	-241	N/A	N/A	N/A

Nomenclature: See separate page